# Nov 11 training: "Inside Power Generation" run of show (draft 1)

**Session:** Wednesday, Nov 11, 2026, 1:30-2:30 PM, on Zoom. About 440 required and optional attendees, mostly internal; they need clarity, not engineering. Host: Stephan Hardt.

**Promised in the invite:** power-generation fundamentals and the market landscape, then a guided virtual tour of a 3D gas power campus (combined cycle, modular generation, BTM, LNG). The tour covers the major equipment and electrical systems, how they work together, and where wire and cable are used.

**The rule for the next 19 days:** if it doesn't serve one of the seven stops below, it goes in the parking lot (end of this file).

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**Slides:** `renders/deck/training/SW-3X1_Nov11_training.pdf` (60 slides, built by `blender/deck_training.py`; JPG copies in `renders/deck/training/jpg/`). See the slide map below.

## Slide map (Oct 25: 58 slides; main deck 1-51, backup 55-58)

| Slides | Section |
|---|---|
| 1-6 | Title, agenda, **what you will take from this hour** (by role), **why a 3D model** (4: V0-V1, 5: V2-V3), **how we built it** (6) |
| 7-10 | Divider 01 *How power is made*; every power plant is this; four families; the mix today and why gas |
| 11-17 | Divider 02 *The market*; at a glance, load growth, backlogs, the cable prize, register, state heat map |
| 18-37 | Divider 03 *Inside a gas plant*; open the box, a day on the grid, the power block snapshot (21), the combined-cycle build (22-25), the X-ray (26 copper, 27 by type), generator to light switch, voltage ladder, cable 101, what a plant does to a cable, owners, greenfield/brownfield, who buys, **three buckets of cable (35)**, **where every cable starts (36)**, the modern campus |
| 38-46 | Divider 04 *The tour*; campus overview; the seven stops |
| 47-53 | Divider 05 *Wire and cable*; the whole campus in one number; **the offer**; **installed faster (SIMpull)**; **knowledge check (Zoom poll)**; take-aways and Q&A; **resources and who to call** |
| 54-60 | Backup: divider, quiz answers, four ways to burn gas, how an HRSG works, follow the power, the HRSG spec, sources and qualifiers |

**New slides, talking points** - **Slides 4-5, why a 3D model (1.5 min):** 4: V0 (March) the first sketches, cable colour-coded but generic (11 kV / 138 kV, no counts); V1 (March) a Southwire-branded concept, hall and HRSGs in detail, still invented. 5: V2 (spring-summer) 3D studies, real elements but not to scale or routed; V3 (September-today) drafted as the SK-3X1 drawing set, then the model built from it: LOD 3, the cable schedule, the X-ray, the tour. Why: see the whole plant, measure instead of guess, one source, speak the buyer's language. - **Slide 6, how we built it (1 min):** research (700+ pages of real combined-cycle drawings: layouts, one-lines, equipment and load lists, to confirm equipment, voltages and electrical systems) → the 16-sheet SK-3X1 drawing set → the 3D model (1,023 items, 156,247 parts, 719 routes) → checks (211/211 footprints, 21 heights, 1.5 M clash pairs, none) → the cable schedule (21,217 cables, 6,011 runs) → one model behind everything. "Conceptual, not engineered, but

not invented." CONFIRM before Nov 11: what the 700+ pages were and whether to name the source. - **Slide 3, what you will take (1 min):** one line per audience: sales, customer service, supply chain and plants, marketing and leadership. "No engineering background needed." - **Dividers 7, 11, 18, 38, 47 (5 s each):** say the section name and move on. - **Slide 35, three buckets of cable (1.5 min):** follow the cable, not the building. 1 inside the packages: OEMs, e-house and package builders buy it, every month, not in our 440 miles. 2 between the packages: the EPC or its electrical contractor, once per plant, most of the miles. 3 to the grid: the HV contractor or utility, early, few cables, high value. "Starting in the e-house is not the same as being bought by the e-house builder. Cover both." - **Slide 36, where every cable starts (1 min):** the R1 switchgear room from above, with the switchgear, MCCs and drives, and the control room. "Power leaves the switchgear and MCCs; signals come home to the control room. Every device in between is wired." Ties the three buckets to a real room. - **Slide 49, the offer (1.5 min):** our families by plant area, grid to thermocouple, plus the services (spec support, cut-to-length, harness kits, logistics, field support). It replaces the HRSG-only spec slide (now backup 59). - **Slide 50, installed faster (1 min):** SIMpull reels (one pull per feeder), SIMpull truck, cut-to-length, pre-terminated. "Every set-up and termination saved is schedule saved." - **Slide 51, knowledge check (2 min):** launch the Zoom poll with the three questions, show results, give the answers by voice (backup 55 has them with the why). Answers: C instrumentation; C the EPC or package builder; B about one-third. - **Slide 53, resources (30 s, leave it up during Q&A):** the viewer link, the brochures, the recording; the team: Stephan Hardt, Patrick Ritchie (EPC accounts), Greyson (EPC outreach), Phil Metz and Bob Benish (OEMs and package builders). - **Slide 60, sources:** for questions on any number.

## Rundown (60 min)

| Time | Segment | On screen | Mins |
|---|---|---|---|
| 1:30 | 0. Welcome and why this matters | 6 slides (title, agenda, why you, why a 3D model x2, how we built it) | 2 |
| 1:32 | 1. Market and fundamentals | 30 slides | 28 |
| 2:00 | 2. The 3D tour: seven stops | Live viewer, plus one slide per stop if needed | 18 |
| 2:18 | 3. Wire and cable takeaways | 3 slides | 6 |
| 2:24 | 4. Q&A | Viewer on the overview | 6 |

Keep 2 minutes of slack inside the tour. If you run long, shorten stop 7 (LNG) to one minute.

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## 0. Welcome (2 min)

**Say:** - "Every power plant runs on cable. Today we walk through a complete gas power campus, built as a 3D model with a full cable schedule, and look at what each part does and where our products go." - "No engineering background needed. At each stop you'll get three things: what it does, its electrical system, and the cable."

**Housekeeping:** questions in the chat, answered at the end. The recording will be shared.

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## 1. Market and fundamentals (28 min, 30 slides)

Market slides 6-11 come from the Oct 6 PowerGen channel deck. **OEM, owner and EPC names are not on the slides; say them here if useful.** Verify the figures against the primary sources (appendix of the Oct 6 deck) before Nov 11.

**Slide 8: Every power plant is this (1.5 min).** The definitions: source (input: fuel or nature) → conversion (the black box: turbine and generator, engine, panel, fuel cell) → power (output: to the grid through a switchyard, or behind the meter). "Every plant fits this picture; what changes is the source and the machine in the box."

**Slide 9: How electricity is made: four families (1.5 min).** Thermal (heat spins a turbine: gas, coal, nuclear, biomass, geothermal, oil), water and wind (hydro, wind), direct (solar PV, fuel cells: no turbine, DC) and storage (batteries, pumped hydro: shifts power, does not make it). Firm vs variable. "Almost all of them spin a generator; every one of them ends in cable. Today: gas, inside the thermal family." Our campus has gas, engines, aeros and package boilers, the fuel-cell bank and the battery yard.

**Slide 10: The mix today, and why gas (2 min).** US electricity 2025 (EIA): gas 41%, nuclear 18%, coal 17%, wind 11%, solar ~7%, hydro 6%. Planned 2026 additions: 86 GW, mostly solar (43.4) and batteries (24), wind 11.8, gas 6.3. "Southwire cables every one of these. Today we zoom in on gas: it is firm, it is the backbone, and the next wave of data-centre load needs it."

**Slide 12: At a glance (1.5 min).** 378 GW in development (52 GW under construction), ~$1.2B core cable 2026-2035, 386 projects tracked, $50M+ booked in Q3. "Today's goal: know the plant, so you can follow the cable through it."

**Slide 13: Load growth (2 min).** 166 GW peak-load growth by 2030 (about 90 GW data centres); 378 GW gas in development (189 GW tied to data centres); 52 GW under construction; turbine lead times up to 7 years. The chart: the 2000-04 merchant boom built 188 GW (peak 57 GW in 2002); 2015-25 averaged 8.1 GW a year. "Today's pipeline needs a boom-scale build again."

**Slide 14: Turbine backlogs (1 min).** Turbine OEM 1 = GE Vernova (116 GW, firm backlog plus slot reservations, Q2 2026); OEM 2 = Siemens Energy (~69 GW gas-turbine backlog, Q3 FY26); OEM 3 = Mitsubishi Power (35 GW large-frame, Aug 2026). Tiles: +72% Caterpillar power-generation sales Q2 2026; 2.3 GW VoltaGrid modular fleet for Oracle's Texas AI data centres plus 1 GW for Vantage; 500 MW standard modular SGT-800 plant from Siemens Energy and Eaton.

**Slide 15: The cable prize (2 min).** 378 GW to 93.7 GW modeled additions to ~$180B of plant investment to $0.75-1.2B of wire and cable: under 1% of capex, on the critical path. By PO year: $673M named projects, $501M future allowance; peak 2028-29.

**Slide 16: The register (1 min).** 205.8 GW eligible; 68% only announced. Top states TX 55.5, LA 16.3, WV 13.4 GW (AZ is 6.1). Example owners if asked: NRG, CPV, Entergy, Homer City Redevelopment. Named EPCs: Kiewit/TIC, Zachry, Gemma, PCL, Burns & McDonnell; 87% of gross capacity has no confirmed constructor yet.

**Slide 17: Where the build is (1 min).** The state heat map: TX 55.5 GW leads, then LA, WV, IN, OH, PA. Owners: IPPs and behind-the-meter developers mostly build new; utilities reuse old sites (coal-site repowers 41.3 GW).

**Slide 19: Open the box: the physical anatomy (1.5 min).** Back to the black box, now for gas: the six steps from fuel infrastructure to end use, grouped under input, black box and output. "Same physics on every project; what changes is the machine in the box, and who owns it."

**Slide 56: Four ways to burn gas for power (2 min).** Simple cycle (the peaker), combined cycle (the workhorse, our power block), engines (the sprinter) and aeroderivatives (the jet): size, efficiency, start time, how they run. Units: GW, MW, kW; MW is power, MWh is energy.

**Slide 20: A day on the grid (1.5 min).** Illustrative 24-hour demand: baseload all day, solar at midday, combined cycle in the middle, peakers and batteries on the evening peak. "Every start cycles the plant and its cable."

**Slide 21: The power block, as you would see it (30 s).** The aerial in full colour, labelled: gas turbines in the hall, HRSGs, stacks, steam turbine, air-cooled condenser, step-up transformers. "This is the power block. Now watch how it makes power, one stage at a time." Click.

**Slides 22-25: Combined cycle, a four-click build (2.5 min).** The same aerial of our power block, darkened; each click lights one stage on the real equipment, and the flows follow the real routes. 15: the gas turbine (air in, power out to the step-up transformers, about 2/3 of the power). 16: the HRSG catches the ~1,100 °F exhaust and makes steam, no extra fuel. 17: the steam turbine makes about 1/3 more; the condenser returns the water. 18: all lit: "Burn the gas once, use the heat twice": ~60% vs ~40%, 3 + 1, 8,234 cables in the power block and its condenser. (The old flow diagram, boilers slide 6, is the backup.)

**Slide 57: How an HRSG works (1.5 min).** Reuse boilers slide 5: the "What it is" and "Fun facts" box.

**Slide 26: Copper X-ray (30 s).** Same view, one more click after the cycle with the steel made glass and every cable glowing copper. "Take the plant away: the cable stays. About 8,200 cables and 440 miles in the power block and its condenser." Let it land; say nothing for a beat.

**Slide 27: X-ray by type (1.5 min).** "Now sort it by job: five kinds of cable." Only what is in the picture: the power block and its air-cooled condenser, 8,234 cables, about 440 miles. Every route glows by type (each route takes the colour of the type with the most cables on it): green instrumentation dominates. MV/HV 232 cables (28 mi), LV power 1,160 (95 mi), control 1,255 (94 mi), instrumentation 5,484 (202 mi), fibre and fire alarm 103 (21 mi). "The big cables get the attention; the small ones are most of the miles." If asked how: every device in the model gets its own cable, routed on the modelled trays: (route + drops + 20 ft) x 1.1. A conceptual estimate, not an engineered take-off. The whole campus (modular yard, BESS, CCS, LNG) is about 21,000 cables and 1,200 miles: slides 22, 27 and 39 (the whole campus in one number) say "whole campus".

**Slide 28: From the generator to the light switch (1.5 min).** Generator 13.8-25 kV → step-up to 115-765 kV → transmission → substation → 4-35 kV distribution → 480/240/120 V at the customer, with the cable family at each step. "Southwire has a product at every step."

**Slide 29: The voltage ladder (2 min).** Reused from the boiler deck (slide 9, "Levels, ratings and the cable they drive"): 230 kV to the grid, 21 kV at the generator, 13.8 kV to the big motors, 480 V to the small ones, 120 V AC / 125 V DC for controls, then signals. "Every rung has its own cable family."

**Slide 30: Cable 101 (1.5 min).** The eight families: HV, MV, LV power, VFD, control, instrumentation, fibre, grounding. By length in SW-3X1: instrumentation 44%, LV power 20%, control 18%, MV/HV 7%.

**Slide 31: What a power plant does to a cable (1.5 min).** Heat, cycling, electrical noise, hazardous areas, fire and smoke, outdoors, schedule, each with the answer (VFD-rated cable, ARMOR-X MC-HL, SIMpull, pre-terminated). "The wrong cable fails early, and a cable failure can trip a unit."

**Slide 58: Follow the power (1.5 min).** SW-3X1 cable by plant area, from the model's schedule: the black box (turbine hall, HRSGs, cooling, water) holds 44% of the campus's 1,212 miles. The POI is where the plant meets the grid: plant side is owner and EPC scope, past it the utility's. "The grid connection carries the biggest cables; the power block carries the most."

**Slide 32: Same physics, different owners (1.5 min).** IPP = one company per plant + asset + power purchase agreement + customer + investors + market (ERCOT, PJM, MISO). Utilities and co-ops buy like utilities; hybrid IPPs both ways; pure IPPs are our sales focus for competitive thermal power.

**Slide 33: Greenfield and brownfield (1.5 min).** Five kinds of site from the register: greenfield grid-connected 77.0 GW, data-centre / behind-the-meter campus 54.5, coal-site repower 41.3, expansion at an operating plant 21.4, industrial captive and site reuse 11.6. New sites take the whole cable scope; existing sites mean tie-ins, live equipment and outage windows. Plus work on running plants (HRSG module replacement: the boiler deck's outage story). Utilities mostly reuse old sites; IPPs and BTM developers mostly build new.

**Slide 34: Who buys, and when (1.5 min).** Develop → finance → engineer (the owner's engineer specifies the cable) → procure (the EPC or package builder buys it) → build (cable pulls on the critical path) → operate. "Be in the spec early; be ready when the EPC buys."

**Slide 37: A modern gas plant (1 min).** Six things beyond the power block, all in the model: private 5G monopole (massive-MIMO), EV charging and solar carport, the BTM data centre, LNG by sea, hydrogen, carbon capture, each with its own cable.

**Slide 39: The campus at a glance (2 min).** The campus aerial with the LNG terminal and the seven stops. - "This is SW-3X1, a 3x1 combined-cycle plant: three gas turbines, three HRSGs and one steam turbine, with a modular generation yard, a battery yard, a behind-the-meter data centre, LNG supply, and options for carbon capture and hydrogen." - "Conceptual and not engineered, but sized like real plants."

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## 2. The 3D tour: seven stops (18 min, about 3 min each; stop 7 is 1 min)

**How to run it:** open the viewer and press **▶ Nov 11** (top left of the 3D window). It goes full screen with captions (Equipment / System / Cable). Use **→ / space** for the next view, ← to go back, keys **1-7** (or the chapter bar at the top) to jump to a stop, and **Esc** to leave. The tour has 22 views: an opening, 20 across the seven stops, and a closing cable X-ray. Use the same three beats at every stop:

1. **What it does** (1 min)
2. **Electrical system** (1 min)
3. **Where the cable goes** (1-2 min, show the cable X-ray)

### Stop 1: Grid connection and switchyard

- **Does:** sends the plant's power to the 230 kV grid.
- **Electrical:** generator step-up transformers (GSUs) take 21 kV up to 230 kV; then breakers, disconnects, and a control house with protection relays.
- **Cable:**
- HV conductor and 230 kV cable;
- control and relay cable from every breaker to the control house;
- fibre;
- a buried ground grid (bare copper).
- **Line:** "Every breaker out here has a bundle of control cable running back to the control house."

### Stop 2: Turbine hall: gas turbines, generators, steam turbine

- **Does:** three H-class gas turbines and one steam turbine turn generators.
- **Electrical:**
- isolated-phase bus from the generators to the GSUs;
- unit auxiliary transformers feed the 13.8 kV switchgear in R1;
- static starters spin the gas turbines up.
- **Cable:**
- MV-105 to big motors and transformers;
- TC-ER power and control to the lube-oil and hydraulic skids;
- thousands of instrument pairs (vibration, temperatures).
- **Show:** the R1 switchgear room interior (13.8 kV lineups, MCCs).

### Stop 3: HRSG and its e-house (the boiler story; your best material)

- **Does:** turns exhaust heat into steam; the SCR inside cuts NOx.
- **Electrical:**
- the 13.8 kV feed pumps (two of 5 MW each);
- the T-R2A transformer feeds the e-house (480 V switchgear, MCCs, drives, DCS);
- CEMS shelter at the stack.
- **Cable:**
- one HRSG train is **707 cables, about 157,000 ft**, and **about 74 % of them are instrument and thermocouple pairs**;
- ARMOR-X MC-HL at the duct-burner gas skid.
- **Show:**
- the HRSG close-up;
- the e-house cutaway with its 9 callouts;
- the CEMS shelter (boilers slides 7, 9 and 10). Pick **one** of the three if time is short.
- **Close the stop on Boiler Row** (the last view of stop 3; boilers slide 4): the package, utility and electric boilers side by side.
- **Line:** "Three boilers, three cable stories. The package boiler is control cable, the utility boiler is instrument cable, and the electric boiler is MV cable."

### Stop 4: Cooling and water: air-cooled condenser and water treatment

- **Does:** the ACC turns spent steam back into water. 80 fans, no cooling-tower water.
- **Electrical:** the R4 e-house holds four VFD lineups that drive the 80 fan motors.
- **Cable:** VFD-rated cable (ARMOR-X MC-HL VFD / TC-ER VFD) to every fan; water-treatment MCCs.
- **Line:** "Variable-frequency drives mean VFD-rated cable; ordinary cable fails early here."

### Stop 5: Modular generation yard

- **Does:** fast-start, flexible power.
- **What's there:**
- reciprocating engine gensets in halls;
- aeroderivative simple-cycle units;
- two package types side by side: a factory-modular aeroderivative (AX-1, LM2500XPRESS class, ~35 MW, shipped as pre-built modules with plug-in interconnects) and an industrial gas-turbine package (IG-1, Titan 250 class, ~23 MW, one skid);
- fuel-cell and microturbine modules;
- the battery storage yard next door.
- **Electrical:** each unit has its own generator step-up and MV e-house; inverters and power-conversion skids for the battery and fuel cells.
- **Cable:**
- MV collection cable;
- DC cable at the battery and fuel cells;
- prefabricated cable spines installed as modules;
- skid pre-wire.
- **Line:** "Modular means factory pre-wiring, which is our opportunity to sell pre-terminated and cut-to-length cable."

### Stop 6: Behind-the-meter data centre

- **Does:** a ~120 MW data-centre campus powered directly by the plant's modular gensets, with the grid only as backup.
- **Electrical:**
- 13.8 kV to 34.5 kV step-up transformers (3 x 90 MVA);
- double-bus 34.5 kV switchgear;
- a 40 MW / 80 MWh battery to absorb load steps;
- two data halls, each fed 2N through 48 pad-mount transformers and UPS skids.
- **Cable:**
- 35 kV MV cable;
- large LV feeders to the UPS skids;
- fibre everywhere.
- **Line:** "Data centres are why gas is growing again, and they're cable-dense on both A and B sides."

### Stop 7: Fuel: gas yard and LNG

- **Does:** brings fuel in and conditions it: the gas metering and regulating (M&R) station, plus an LNG satellite of 8 tanks holding about 414,000 gal (about 3 hours of the whole plant) for backup.

- **Electrical:** an LNG e-house (13.8 kV / 480 V); pumps, vaporizers and the boil-off compressor.
- **Cable:** this is hazardous-area territory, so ARMOR-X MC-HL (Class I Div 1 and 2) and TC-ER control, PLTC instrumentation and fire alarm on one strut rack.
- **Line:** "Wherever there's gas, the cable has to be rated for it."

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## 3. Wire and cable takeaways (6 min, 3 slides)

**Slide A: The whole campus in one number (3 min)** - The model's cable schedule: **6,011 runs, 21,217 cables, about 6.4 million ft (~1,200 miles).** - By footage: - instrumentation, 2.8 M ft; - 600 V power, 1.3 M ft; - control, 1.1 M ft; - MV, about 0.4 M ft. - Message: "Most of the footage is small cable: instrument and control. MV is the high-value part, but it isn't the volume."

**Slide B: Product map (3 min).** Reuse boilers slide 13, "What the spec calls for", or the brochure product map: - MV-105; - TC-ER; - ARMOR-X MC-HL; - instrumentation and thermocouple extension; - SIMpull THHN; - fibre and fire alarm.

**Slide C: Pain points and our answer (3 min).** Reuse boilers slide 14.

**Close with three things to remember:** 1. Gas plants are growing again, driven by data centres and firm power. 2. Most of a plant's cable is instrument and control; the big MV and power cable is the smaller share. 3. Southwire covers the whole plant: one supplier, staged on SIMpull reels, spec-compliant from MV down to the thermocouple.

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## 4. Q&A (6 min)

**Questions to prepare for:** - "How long does a plant take to build?" 3-4 years for a combined-cycle plant; modular units are much faster. - "Who are the customers?" Utilities, independent power producers, EPCs, boiler and turbine OEMs, and data-centre developers. - "Is the model a real plant?" Conceptual, sized from public filings of real plants, and not engineered. - "What about hydrogen and carbon capture?" Both are in the model as options; show them if asked (layers on).

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## Backup plan

- **Recorded video of the tour:** screen-record the viewer stop by stop by Nov 3. If the viewer lags on Zoom, play the video and narrate over it.
- **PDF of all slides,** in case screen sharing fails.
- **Second laptop or phone** logged in as co-host.

## Checklist

| Date | Task |
|---|---|
| Oct 24-27 | React to this draft; mark what to cut or keep. Tour mode done (▶ Nov 11 in the viewer); walk it once and note any view to change. |

| Date | Task |
|---|---|
| Oct 28-31 | One slide per stop (reuse zone and boiler slides). Optional: modular turbine packages, **only if done in one day**. |
| Nov 2-3 | Full-quality render batch overnight (renders/RENDER_QUEUE.md); record the backup video. |
| **Nov 4** | **Dry run with one colleague, timed.** |
| Nov 5-9 | Fixes only. |
| Nov 10 | Zoom tech check; freeze. |

## Parking lot (after Nov 11)

- Modular yard and Boiler Row are **closed for Nov 11** (Oct 24): no more equipment, fixes only.
- Possible later plant types: a CHP / cogeneration unit (gas turbine + HRSG selling steam to an industrial host), linear generators.
- Wastewater-treatment and cooling-tower refinements.
- Turbine close-ups at HRSG depth.
- More switchgear-room detail.
- Blender fly-through animation.
- Zone 07 parts 3-4 (construction compound, deck slide).
- Modular gas-turbine packages: done Oct 24 (AX-1 LM2500XPRESS class, IG-1 Titan 250 class). A larger SGT-800-class unit was tried and removed: it repeated the SC-1 / SC-2 simple-cycle story.